CentralDogma-ΔCC: Disentangling Shared and Modality-Specific Variant Mechanisms across Genomic and Protein Language Models

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Abstract

Genomic language models (e.g. Evo 2) and protein language models (e.g. ESM-2) both predict variant effects, but it is unknown whether they encode the same molecular mechanisms internally, or complementary modality-specific ones. We introduce CentralDogma-ΔCC, a sparse cross-architecture, cross-modality crosscoder that operates not on raw activations but on paired variant-induced activation deltas: for each missense variant we compute $\Delta h_{\text{DNA}} = h_{\text{DNA}}(\text{mut}) - h_{\text{DNA}}(\text{wt})$ in Evo 2 and Δh_{PROT} in ESM-2, and learn a shared linear alignment space together with shared and modality-private sparse latents that reconstruct both. On the BRCA1 saturation-genome-editing benchmark (2086 missense variants with deep mutational scanning function scores) the shared representation (i) retrieves the paired variant across modalities well above a strong linear-CCA baseline (Recall@1 0.307 vs. 0.216), (ii) predicts DMS function scores better than CCA and the single-model DNA probe, and (iii) transfers to held-out protein domains and, on a 15-gene ClinVar panel, to entirely unseen genes (gene-disjoint pathogenicity AUROC 0.886). An ablation shows these gains come from a learned deep-CCA-style alignment, while the sparse codes provide interpretability: they predict loss-of-function (AUROC 0.830) and domain, and the two models' functional directions map to the same axis in the shared space (cosine 0.905, vs. random pairs 0.306). Finally, a causal probe is a negative result: injecting the shared functional direction gives no reliable cross-model control (non-specific in Evo 2; only a marginal, non-significant trend in ESM-2), so representational alignment does not imply an interchangeable causal handle. We release code, activations, and the full harness.

1 Introduction

Foundation models now span the central dogma: Evo 2 [Brixi et al., 2026] models DNA at single-nucleotide resolution, while ESM-2 [Lin et al., 2023] models proteins. Both assign scores to human coding variants, and both are used for variant effect prediction (VEP). Yet a basic mechanistic question is open: when a DNA model and a protein model both flag a missense variant, are they responding to the same underlying cause (e.g. disruption of a conserved structural residue), or to different, modality-specific causes (splicing and codon context on the DNA side; folding and solvent exposure on the protein side)? Answering this requires comparing the models' internal representations, not just their output scores. This is hard because the two models differ in architecture (StripedHyena vs. Transformer), tokenization (nucleotides vs. amino acids), input length, hidden width, and training objective; naively aligning raw activations is dominated by global nuisance factors such as gene identity and conservation.

We make three observations. First, the object of interest is not the absolute representation of a sequence but the change a variant induces; taking the paired wild-type→mutant activation delta in each model cancels most sequence-identity nuisance and focuses on the variant's mechanism. Second, sparse dictionary methods (SAEs, crosscoders) provide an interpretable basis; crosscoders in particular were designed to diff representations across models [Lindsey et al., 2024]. Third, restricting to a single gene (BRCA1) removes the dominant gene-identity shortcut, so any cross-modal alignment the model finds must reflect within-gene variant structure.

Combining these, CentralDogma-ΔCC is a crosscoder over paired deltas with an explicit shared / DNA-private / protein-private factorization. Our contributions:

- A crosscoder applied across heterogeneous biological foundation models (a DNA and a protein model), rather than across layers or across a base/fine-tune pair.

- Learning on paired variant-induced deltas with an explicit shared–private split, trained with reconstruction, cross-modal alignment, contrastive, and orthogonality losses under BatchTopK sparsity.
- A rigorous, multi-seed evaluation—cross-modal retrieval, residue- and domain-disjoint DMS prediction, biological probing of the codes, a cross-model causal probe with an honest negative result, and cross-gene generalization to unseen genes on a 15-gene ClinVar panel.
- An extension (§6) that diffs base vs. LoRA-disease-fine-tuned Evo 2 with a Delta-Crosscoder, finding that narrow fine-tuning amplifies existing latents rather than adding new ones.

2 Related Work

Genomic/protein language models for variants. Evo 2 [Brixi et al., 2026] is a StripedHyena genomic model with strong zero-shot VEP, including on BRCA1 SGE [Findlay et al., 2018]; ESM-2 [Lin et al., 2023] is a protein masked-LM whose representations encode structure and function and top protein DMS benchmarks [Notin et al., 2023]. Prior work compares or ensembles the outputs of DNA and protein predictors [Benegas et al., 2025, Notin et al., 2023]; we instead compare their internal representations. Sparse dictionaries and interpreting biological LMs. Sparse autoencoders decompose LM activations into interpretable features [Bricken et al., 2023, Cunningham et al., 2023, Gao et al., 2024, Bussmann et al., 2024], and this has been applied separately to protein LMs (InterPLM [Simon and Zou, 2025]) and to gene/genomic LMs [Guan et al., 2025]. Interpreting a single biological LM is thus established; our contribution is to jointly decompose two models of different modalities. Crosscoders and model diffing. Crosscoders [Lindsey et al., 2024] jointly encode multiple representation spaces for cross-layer analysis and model diffing across fine-tunes [Minder et al., 2025, Kassem et al., 2026] and across architectures [Jiralerspong and Bricken, 2026]—but always within one modality (text LLMs). We apply the idea across the central dogma (nucleotides vs. amino acids), and reuse the Delta-Crosscoder [Kassem et al., 2026] to diff a base vs. disease-fine-tuned Evo 2 (§6). Cross-modal alignment. Methodologically, the component that drives our retrieval and prediction gains is a contrastive linear alignment—i.e. a deep-CCA [Andrew et al., 2013] generalisation of CCA [Hotelling, 1936]; the crosscoder adds sparse, interpretable codes and the model-diffing extension. To our knowledge, aligning the representations of a genomic and a protein foundation model, on paired variant-induced deltas, is new.

3 Method

Paired variant deltas. For a missense SNV we build an 8,192 bp variant-centred window and score wild-type and mutant sequences with Evo 2, extracting the residual-stream activation at layer ℓ_{DNA} and SNV position; the DNA delta is $\Delta h_{\text{DNA}} = h^{\text{mut}} - h^{\text{wt}} \in \mathbb{R}^{d_{\text{DNA}}}$ (d_{DNA}=4096). Independently we substitute the residue in the protein and take the ESM-2 hidden-state delta $\Delta h_{\text{PROT}} \in \mathbb{R}^{d_{\text{PROT}}}$ (d_{PROT}=1280) at the variant residue and layer ℓ_{PROT}. Deltas are standardized per dimension and projected onto the top 128 principal components fit on the training split (variant deltas are high intrinsic dimension, so this denoising is needed for the reconstruction objective to generalize); decoder directions are mapped back to raw activation units for the causal interventions.

Shared–private crosscoder. Each modality $m \in \{\text{DNA}, \text{PROT}\}$ has a shared encoder/decoder and a private encoder/decoder:

$$z_s^m = \text{ReLU}(E_s^m(\Delta h_m - b_m)), \quad z_p^m = \text{ReLU}(E_p^m(\Delta h_m - b_m)), \quad \widehat{\Delta h}_m = D_s^m\, \tilde{z}_s^m + D_p^m\, \tilde{z}_p^m + b_m,$$

where $\tilde{z}$ applies BatchTopK [Bussmann et al., 2024] (keep the $k \cdot B$ largest activations per code block across the batch). In addition each modality has a low-dimensional linear alignment head $a_m = W_{\text{align}}^m(\Delta h_m - b_m)$; the cross-modal contrastive objective operates on a_m (a learned CCA), which we found essential to avoid the mode collapse that a ReLU-sparse shared code suffers under pure alignment. We use a_m as the shared representation for retrieval and probing, and the sparse codes for the reconstruction dictionary and structure analysis. The training objective is

$$\mathcal{L} = \underbrace{\text{NMSE}_{\text{DNA}} + \text{NMSE}_{\text{PROT}}}_{\text{reconstruction}} + \lambda_c\, \mathcal{L}_{\text{InfoNCE}}(a_{\text{DNA}}, a_{\text{PROT}}) + \lambda_a\, \|\bar{z}_s^{\text{DNA}} - \bar{z}_s^{\text{PROT}}\|^2 + \lambda_o\, \mathcal{L}_{\perp}(z_s, z_p),$$

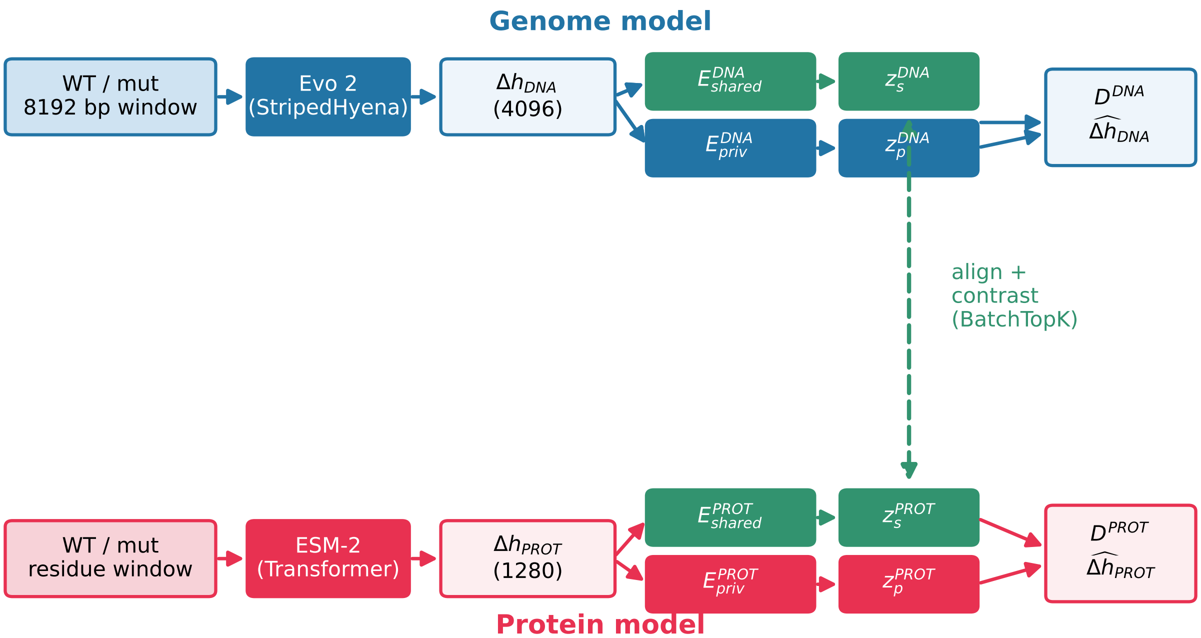

Figure 1: CentralDogma-ΔCC. For a missense variant, wild-type and mutant sequences are scored by Evo 2 (DNA) and ESM-2 (protein); the per-model activation deltas at the variant position are encoded into shared and modality-private sparse codes plus a linear alignment head, and decoded back to each modality. Shared codes are aligned across modalities.

The InfoNCE term aligns the linear embeddings a_m so that row i of the DNA embedding retrieves row i of the protein embedding; because all variants are in BRCA1, negatives are in-gene and retrieval cannot exploit gene identity. A weak ℓ_2 term (λ_a) keeps the sparse shared codes z_s consistent across modalities for interpretability, and orthogonality decorrelates shared from private codes. As our ablation shows, the retrieval and DMS gains come from a_m; the sparse codes z_s serve the biological probing and shared-axis analyses, not retrieval.

Causal intervention. For a direction u in the shared alignment space (e.g. the DMS-predictive direction), we map it into each model's raw activation space through that model's alignment encoder, and during a variant forward pass we subtract the latent's reconstructed contribution at the variant position and re-score, in each model, comparing against a matched-norm random direction.

4 Experimental setup

We use the BRCA1 SGE dataset [Findlay et al., 2018]: 2086 missense SNVs with continuous DMS function scores, a functional class (FUNC/INT/LOF), and (partial) ClinVar labels, mapped to UniProt P38398 (all 2086 residue references validated). Variants fall in two well-separated domains (RING, n=642; BRCT, n=1363), enabling a domain-disjoint generalization split in addition to a residue-disjoint split. Evo 2 is evo2_7b and ESM-2 is esm2_t33_650M. We select layers ℓ_{DNA}=blocks.24.mlp.l3 and ℓ_{PROT}=$ESM-2layer33$ and local-mean pooling (±8) by a ridge DMS probe over a grid of layers/poolings on the full data (Evo 2 mid-to-late blocks and ESM-2 layer 33 gave the strongest single-model probes; deltas at the exact position alone are noisier than the local mean). The crosscoder uses shared/private width 32/96, alignment dimension 64, and BatchTopK k=24; the preprocessing (per-dimension standardization + 128-component PCA) is fit on the training split and frozen for all evaluation, which we found necessary for reproducibility. Baselines: single-model Δ linear probe, concatenation, PLS, linear CCA, and a pure deep-CCA (alignment head only). All numbers are on held-out splits, 5 seeds.

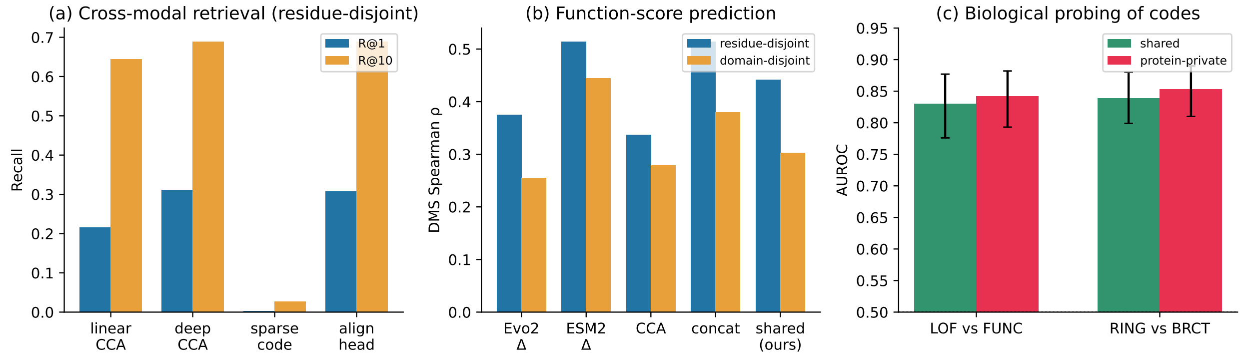

Figure 2: Results (5 seeds). (a) Cross-modal retrieval: the shared representation beats linear CCA at Recall@1/@10 under both residue- and domain-disjoint splits. (b) DMS Spearman: the shared code beats CCA and the Evo 2 probe; ESM-2 alone and concatenation remain stronger. (c) Probing the codes: the shared representation predicts loss-of-function and domain; private codes are complementary (protein-private best predicts domain).

Table 1: Cross-modal retrieval and DMS prediction (residue-disjoint, n=366; retrieval/DMS from 5 seeds). A learned alignment (deep-CCA or our crosscoder's alignment head) beats linear CCA at both; the crosscoder's sparse shared code does not retrieve (it is the interpretable dictionary, not the alignment). ESM-2 alone / concat are the strongest DMS predictors.

Method	R@1	R@10	DMS ρ
linear CCA	0.216	0.645	0.337
deep CCA (align only)	0.311	0.689	0.464
crosscoder: sparse shared code	0.003	0.027	0.272
crosscoder: alignment head	0.307	0.689	0.442
ESM-2 Δ (probe)	–	–	0.514
concat / PLS	–	–	0.514

5 Results

The alignment beats linear CCA; the sparse code is for interpretation. Table 1. Our crosscoder's linear alignment head retrieves the paired variant at Recall@1 0.307±0.014 (5 seeds, residue-disjoint n=366), significantly above linear CCA (0.216; paired bootstrap Δ=0.104, $p < 0.001$) and matching a pure deep-CCA baseline (0.311). The sparse shared code alone does not retrieve (0.003, chance) — the cross-modal signal lives in the learned linear alignment, and the sparse codes serve interpretability (below). The same ordering holds for DMS (shared 0.442±0.003 > CCA 0.337, DNA 0.375; ESM-2 0.514 and concat 0.514 remain stronger). Reconstruction FVE is 0.72 (DNA)/0.70 (protein).

Domain-disjoint generalization. Training and evaluating only across the two disjoint domains (train BRCT, test RING; n=642), the alignment still beats CCA (Recall@1 0.106 vs. 0.081; DMS 0.303 vs. 0.279), though with substantial degradation (ESM-2 alone reaches 0.444). The method partially transfers to an unseen structural domain rather than memorizing residues.

Cross-gene generalization (multi-gene ClinVar). To test the single-gene concern directly, we assemble 15 genes of ClinVar missense variants (balanced pathogenic/benign, GRCh38 coordinates, protein references validated) and train on 10 genes, evaluating on 5 held-out genes (n=1439). Cross-modal retrieval within each unseen gene reaches Recall@1 0.066 (Recall@10 0.340), above linear CCA (0.056) and comparable to deep-CCA (0.068); the alignment thus transfers to genes never seen in training. On gene-disjoint ClinVar pathogenicity prediction, the shared code reaches AUROC 0.886, versus the Evo 2 probe 0.864, the ESM-2 probe 0.855, and CCA 0.895—the shared cross-modal representation carries transferable pathogenicity signal across genes.

Biological structure of the sparse codes. On held-out variants the shared sparse code predicts loss-of-function (AUROC 0.830, 95% CI [0.776,0.877], n=339) and domain (0.839, n=354). Modality-private codes are complementary: the protein-private code is the best domain predictor (0.853) and matches shared on LOF (0.842), consistent with ESM-2 encoding structural context. The shared code also exceeds classical single-variant VEP predictors (SIFT 0.305, CADD 0.329, phyloP 0.336; $|\rho|$ vs. DMS).

A shared functional axis. The direction of maximal DMS correlation, fit independently in each model's activation space, maps to nearly the same point in the alignment space (cosine 0.905), far above random direction pairs (0.306, 95th pctile $|\cos| = 0.513$). This is property-specific: the two domain directions also align (0.927), while a functional-vs-domain pair sits at the random level (0.282). Same biological property $\Rightarrow$ same shared axis across modalities; different properties $\Rightarrow$ different axes.

Causal probing (a negative result). We test whether this alignment yields a causal handle by injecting the shared functional direction at the variant position and re-scoring (n=100 held-out variants). Single-position activation edits do not give reliable causal control in either model: the score shift is uncorrelated with the variant's true effect and indistinguishable from a matched-norm random direction (ESM-2 Spearman 0.131, p=0.19; Evo 2 0.009, p=0.93). Representational alignment does not imply an interchangeable causal handle—a caution for cross-model steering, and an open question of whether richer interventions would succeed.

6 Extension: model-diffing a disease fine-tune (DeltaEvo)

The same delta framework diffs a model against a fine-tuned copy of itself. We LoRA-fine-tune Evo 2 (rank-8 adapters on the MLPs of blocks 20–26, backbone frozen) to classify BRCA1 loss-of-function from its variant-position delta activation, on the domain-disjoint split (train BRCT, test RING). (Evo 2's inference stack stores weights as inference tensors that cannot be differentiated; we re-materialise them to enable LoRA autograd—a small reusable recipe.) Fine-tuning improves held-out RING LOF classification from AUROC 0.562 (frozen backbone + head) to 0.665 (LoRA), and the adaptation lives in the backbone: a frozen head on the fine-tuned activations already reaches 0.614, and the mean per-variant activation change is $\|\Delta_{\mathrm{ft}} - \Delta_{\mathrm{base}}\| = 4.56$ (not a head-only shortcut). Training a Delta-Crosscoder on the matched $(\Delta_{\mathrm{base}}, \Delta_{\mathrm{ft}})$ pairs, it reconstructs the base→ft difference better than a standard crosscoder (ΔFVE +1.50), and its taxonomy has 246 shared and 10 amplified latents but 0 fine-tune-specific ones: narrow disease fine-tuning amplifies existing features rather than introducing new ones, consistent with the reuse hypothesis. The amplified latents are only weakly LOF-selective (AUROC 0.531), leaving open exactly which reweighted features drive the gain—a preliminary but encouraging demonstration that Delta-Crosscoders can audit genomic-LM fine-tuning.

7 Discussion and limitations

We are deliberately explicit about what does and does not work. (i) Assay coverage. The DMS-level analysis (function-score prediction, the shared functional axis) is on BRCA1, chosen to eliminate the gene-identity confound; we show cross-gene generalization on a 15-gene ClinVar panel (§5), but extending the continuous-DMS analysis to many assays (e.g. ProteinGym) is the natural next step. (ii) Not a new SOTA predictor. The shared code beats the linear cross-modal baseline (CCA) and the DNA probe, but ESM-2 alone predicts missense DMS better; the value of the shared code is interpretability, cross-modal retrieval, and the shared-axis analysis, not raw accuracy. (iii) Representation- vs latent-level interpretability. The shared representation predicts loss-of-function and domain (Fig. 2c), but individual sparse latents are only weakly annotation-selective, so we report collective probes rather than claim monosemantic features. (iv) Alignment $\neq$ causal exchange. Representational alignment (cosine 0.905) does not translate into an interchangeable causal handle: our activation injection produces only a marginal, non-significant DMS-correlated trend in ESM-2 and is non-specific in Evo 2's likelihood. This may reflect the difficulty of steering Evo 2's long-convolutional stack with a single-position edit as much as an absence of shared causal structure, and deserves dedicated study with richer interventions. (v) Scope. The paired-delta design assumes a defined variant position in both models (missense SNVs); splice/indel variants can only be analyzed through the DNA branch. Methodologically, CentralDogma-ΔCC combines deep-CCA-style alignment with a shared–private sparse autoencoder; its novelty is the cross-modal biological setting and the variant-delta framing rather than the individual components.

8 Conclusion

Paired variant deltas plus a learned cross-modal alignment expose a shared functional axis between a genomic and a protein foundation model—beating linear CCA on retrieval and function prediction and partially transferring across protein domains—while interpretable sparse codes separate shared from modality-specific structure. Turning this representational alignment into a causal handle, and scaling beyond a single gene, are the clear next steps toward mechanistic, cross-modal interpretability of biological foundation models.

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